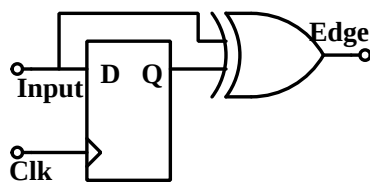
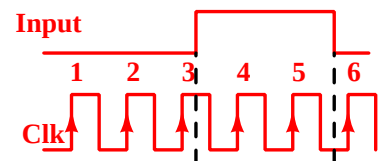


## Tutorial-12 Sequential Circuits

Q-1

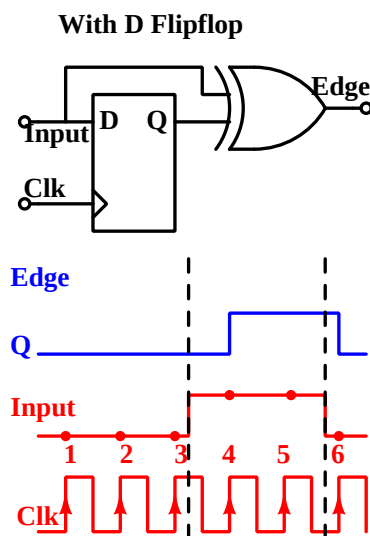


You are analyzing the circuit shown on the left, which feeds an Input Signal (shown on the right) into a D FF and then compares that Input Signal with the output (Q) using an XOR gate.



- a) Draw the output of the XOR gate (Edge) for the given Input signal and Clk. Comment on its functionality.

Soln. 1-a)



The input is sampled as '0' on the first three positive edges of the clock. At the fourth edge, a '1' is sampled, so the output goes to '1' here. At the fifth edge, a '1' is sampled again, so the output remains at '1'. At the sixth edge, a '0' is sampled, so the output returns to '0'.

Till the middle of the positive cycle after the 3rd clock, D and Q are both '0', so the output of the XOR gate (Edge) remains '0'.

In the middle of the positive cycle after the 3rd clock, Input rises to '1', but Q is still '0'. Since D and Q differ, Edge = '1'. At the fourth clock, Q also becomes '1'. D and Q are now equal, so Edge = '0'.

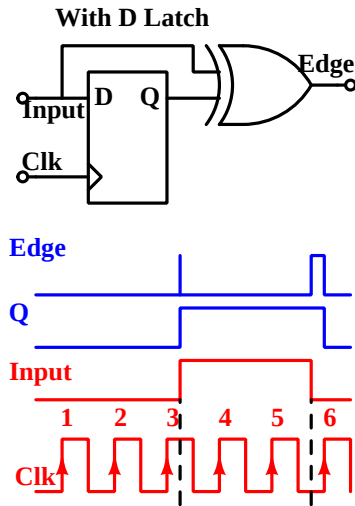
During the low phase of the clock before the 6th positive clock edge, the input becomes '0', while the output is still '1'. So Edge = '1'.

At the sixth positive clock edge, the output drops to '0' and the output becomes '0' again.

Thus Edge outputs a pulse at either edge of the input.

- b) Now, assume you replace the Flip-Flop with a D Latch, and you connect the Clk signal to the Enable Pin of a D Latch. Draw the output of the XOR gate and comment on its differences from part A.

Soln. 1-b)



Since we are using a D latch now instead of a flipflop, the output follows D in the positive phase of the clock. Thus, the output becomes '1' as soon as the input goes to '1' during the positive phase of the clock after the third positive clock edge. Since the inputs of the XOR gate remain equal, its output is '0'.

(Due to delay through the latch, there may be a transient spike to '1').

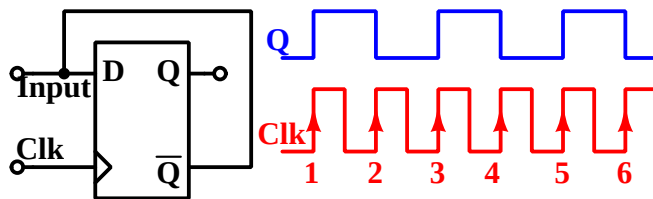
The input returns to '0' during the negative phase of the clock, so Q does not follow D. At this time D and Q are unequal, so the XOR output goes to '1'. When the sixth positive clock edge arrives, Q becomes equal to D and the XOR output drops to zero.

If a latch is used, the output (Edge) detects only the transition which occurs when the clock is '0'.

**Q-2** You have a single positive-edge triggered D FlipFlop. You wire the inverted output ( $\overline{Q}$ ) directly back into the D input. If you connect a clock signal (as shown in Question 1) to the Clk pin, draw the output Q of the DFF.

Comment on the frequency of CLK and obtained Q signal.

**Soln. 2)** Since  $\overline{Q}$  is connected to D, the flipflop will change stage at every positive edge of the clock. (Acts like a T flipflop).



Every time a positive edge occurs on the clock, Q output becomes  $= \overline{Q}$ . Since two transitions are needed per cycle, the output goes through 1 cycle while the clock goes through 2 cycles. Thus the circuit acts like a frequency divider.

**Q-3** You are given the conversion table as shown here, to build the D Flip Flop using an S-R FF.

a) Fill the required S and R inputs based on  $Q_n$  and  $Q_{n+1}$ .  
(i.e. Excitation Table for SR FF).

**Soln. 3-a)** An SR FF has the same behaviour as an SR latch, except that the output changes only on the active edge of a clock signal.

| Input<br>D | Present State<br>$Q_n$ | Next state<br>$Q_{n+1}$ | S | R |
|------------|------------------------|-------------------------|---|---|
| 0          | 0                      | 0                       | 0 | 1 |
| 0          | 1                      | 0                       | 0 | 1 |
| 1          | 0                      | 1                       | 1 | 0 |
| 1          | 1                      | 1                       | 1 | 0 |

Using an SR FF (not latch), the design is quite trivial, since the next state  $= D$  and does not depend on the previous stage  $Q_n$ .

b) Use K-maps to find S and R expression in terms of D and  $Q_n$ .

**Soln. 3-b)** Actually, we do need Karnaugh maps!

We can see that  $S = D$  and  $R = \overline{D}$ .

So we can connect S to D directly and R to D through an inverter.  
(Recall that this solution did not work when we used it with an SR **latch**).

c) Draw the resultant logic circuit using SR FF.

**Soln. 3-c)** Just connect S to D and R to the output of an inverter whose input is D.

d) Derive the expressions for S and R if we want to implement a transparent D latch using an SR **latch** with an enable input E.

**Soln. 3-d)** The truth table now becomes:

| E | D | $Q_n$ | $Q_{n+1}$ | S | R |
|---|---|-------|-----------|---|---|
| 0 | 0 | 0     | 0         | 0 | X |
| 0 | 0 | 1     | 1         | X | 0 |
| 0 | 1 | 0     | 0         | 0 | X |
| 0 | 1 | 1     | 1         | X | 0 |
| 1 | 0 | 0     | 0         | 0 | 1 |
| 1 | 0 | 1     | 0         | 0 | 1 |
| 1 | 1 | 0     | 1         | 1 | 0 |
| 1 | 1 | 1     | 1         | 1 | 0 |

When  $E = 0$ ,  $Q_{n+1} = Q_n$

If  $Q_n = 0$ ,  $S = 0$ , R can be 1 or 0.

if  $Q_n = 1$ ,  $R = 0$ , S can be 1 or 0.

This leads to the following K maps:

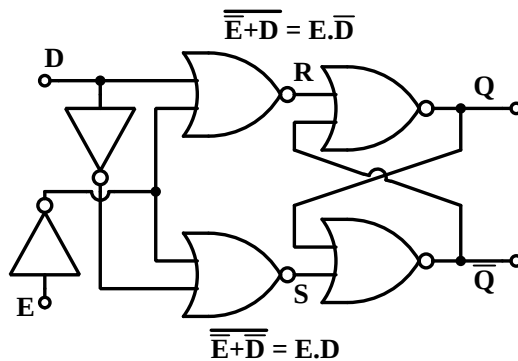
**For S**  $D, Q_n \rightarrow$

|     | 00 | 01 | 11 | 10 |
|-----|----|----|----|----|
| E 0 | 0  | X  | X  | 0  |
| E 1 | 0  | 0  | 1  | 1  |

**For R**  $D, Q_n \rightarrow$

|     | 00 | 01 | 11 | 10 |
|-----|----|----|----|----|
| E 0 | X  | 0  | 0  | X  |
| E 1 | 1  | 1  | 0  | 0  |

So we get  $S = E.D$ ,  $R = E.\overline{D}$ . The circuit diagram below shows how this can be implemented using only 2 input NOR gates and an inverter.



**Home exercise:** Do this problem if the SR latch is implemented as cross connected NANDs, whose inputs are  $\overline{S}$  and  $\overline{R}$

**Q-4** You are designing a logic circuit for a simple cargo elevator that serves only two floors: Floor 0 (Ground) and Floor 1 (Top).

State  $Q=0$  means the elevator is at Floor 0. State  $Q=1$  means the elevator is at Floor 1. There is a single "Move" button, labeled M. If  $M=0$ , the elevator stays where it is. If  $M=1$ , the elevator moves to the other floor.

- Draw the state transition diagram (circles for states, arrows for transitions based on M).
- Fill out the State Transition Table.
- Derive the Boolean logic equation for the Flip-Flop input D in terms of current state Q and input M. What basic logic gate does this represent?

**Q-5** Design a simple 2-bit synchronous counter that continuously counts upward:  $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$ , using D Flip-Flops.

- Write the State Transition Table showing Current State ( $Q_1, Q_0$ ) and Next State ( $D_1, D_0$ ).
- Find the simplified Boolean logic equation for  $D_0$ .

- c) Find the simplified Boolean logic equation for D1.

**Q-6** The campus has installed a new automated Chai dispenser. A cup of chai costs exactly 20 Rupees. The machine only accepts 10-Rupee coins.

- a) To save hardware costs, your institute decides to install an “instant” dispenser. The exact moment the second 10-Rupee coin is inserted, it immediately pours the chai and resets. It never “waits” with 20 Rupees inside.

Input X:  $X=1$  for exactly one clock cycle when a 10-Rupee coin is inserted. ( $X=0$  otherwise).

Output Y:  $Y=1$  triggers the dispenser to pour the chai ( $Y=0$  keeps it locked).

Use exactly one D Flip-Flop. State 0 ( $Q=0$ ) means 0 Rupees inserted. State 1 ( $Q=1$ ) means 10 Rupee inserted.

1. Draw the State Transition Diagram (using arrows labeled X/Y).
  2. Fill out the State Transition Table.
  3. Derive the simplified logic equations for the Next State (D) and Output (Y).
  4. Sketch the final gate-level circuit.
- b) Students complained that the “instant” dispenser from Part A spilled chai on their hands because it poured too fast! You are tasked with upgrading the machine. The machine must now hold the 20 Rupees and wait for the user to press a button before pouring.
- Input X=1 when a 10-Rupee coin is inserted,  
P=1 when the “Pour” button is pressed. (Assume X and P are never pressed at the exact same time).
- Output Y:  $Y=1$  pours the chai.
- The machine now requires three states:  
S0 (00): 0 Rupees, S1 (01): 10 Rupee, S2 (10): 20 Rupees (Ready to pour), and  
S3 (11): Invalid / Don’t Care.
1. Draw the new State Transition Diagram. Include states S0, S1, and S2. Show what happens if X is inserted, and what happens if P is pressed. (Hint: If P is pressed while in S0 or S1, the machine should just stay in that state).
  2. Create the State Transition Table. There are 2 Flip-Flops ( $Q_1, Q_0$ ) and 2 External Inputs (X, P).
  3. Derive the logic equations for the Next States (D1, D0) and Output (Y).

(Optional Practice Question)

**Q-7** Design a synchronous BCD (Binary-Coded Decimal) up-counter using positive-edge triggered D Flip-Flops. The counter must sequence from 0000 to 1001 and roll over to 0000 on the next clock cycle.

- a) Construct the complete table detailing Current State ( $Q_3, Q_2, Q_1, Q_0$ ) and Next State ( $D_3, D_2, D_1, D_0$ ). Explicitly define states 1010 through 1111 as “don’t care” (X) conditions.
- b) Draw and solve four separate 4-variable K-maps to derive the inputs for D3, D2, D1, and D0. Show all your groupings.

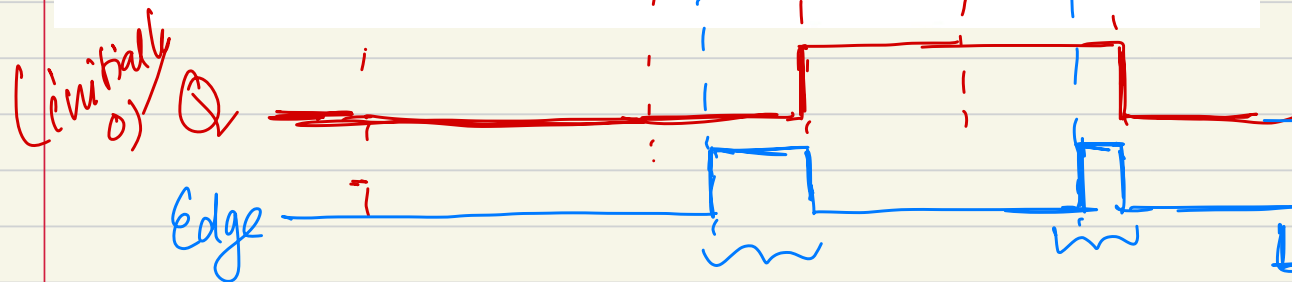
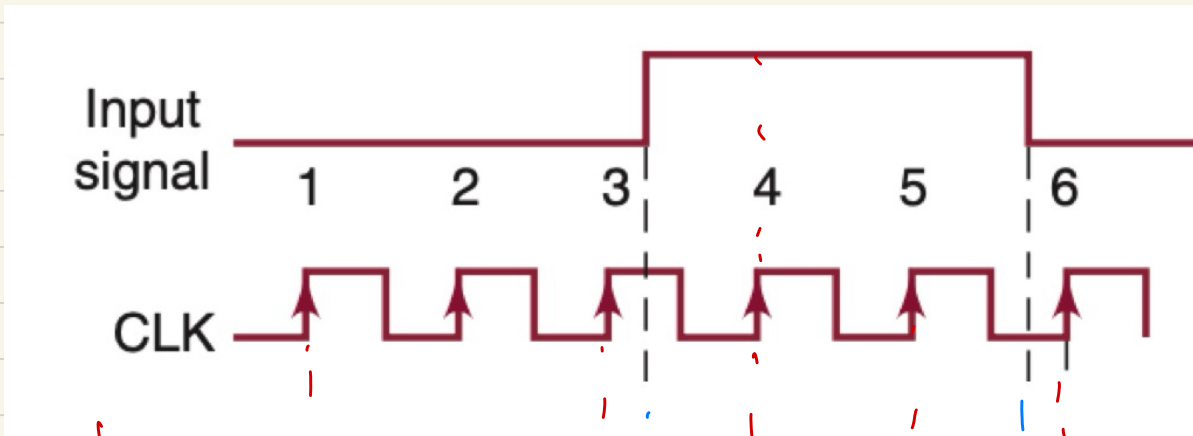
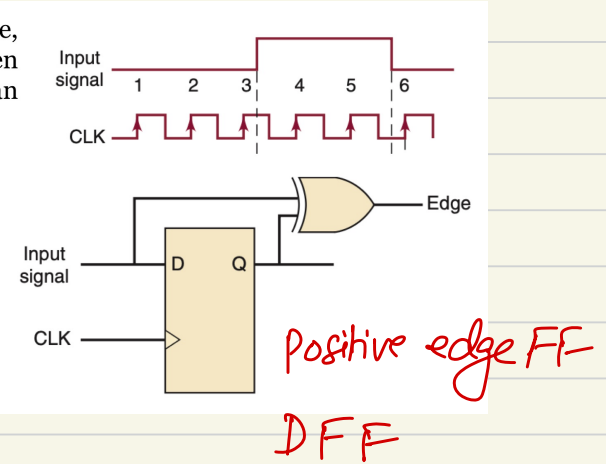
- c) Provide the final, maximally simplified Boolean sum-of-products equations for all four Flip-Flop inputs.
- d) Sketch the final hardware implementation showing the four D Flip-Flops, the clock routing, and the combinational logic gates.

# Tutorial 12

## Solutions

Q1. You are analyzing the circuit shown in the figure, which feeds an Input Signal into a D FF and then compares that Input Signal with the output (Q) using an XOR gate.

- Draw the output of the XOR gate (Edge) for the given Input signal and CLK. Comment on its functionality.
- Now, assume you replace the Flip-Flop with a D Latch, and you connect the CLK signal to the Enable Pin of a D Latch. Draw the output of the XOR gate and comment on its differences from part A.

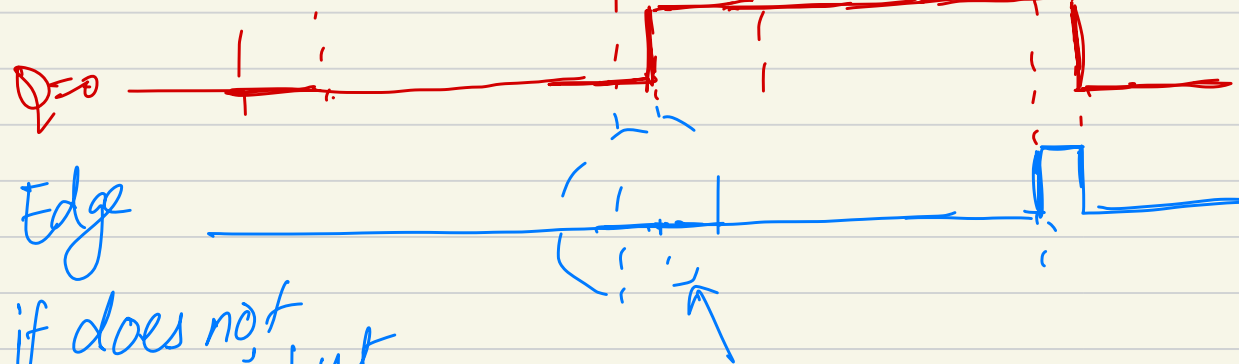
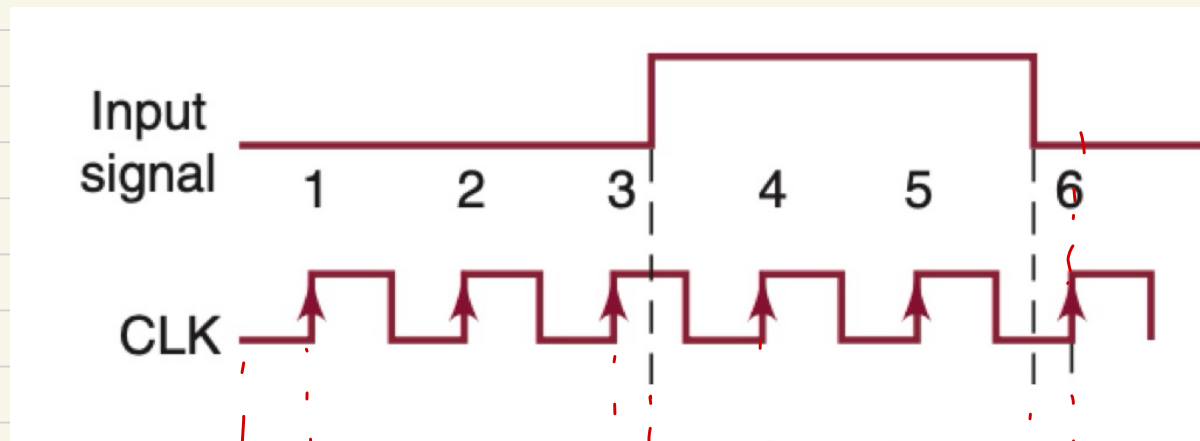


$$\text{Edge} = D \oplus Q$$

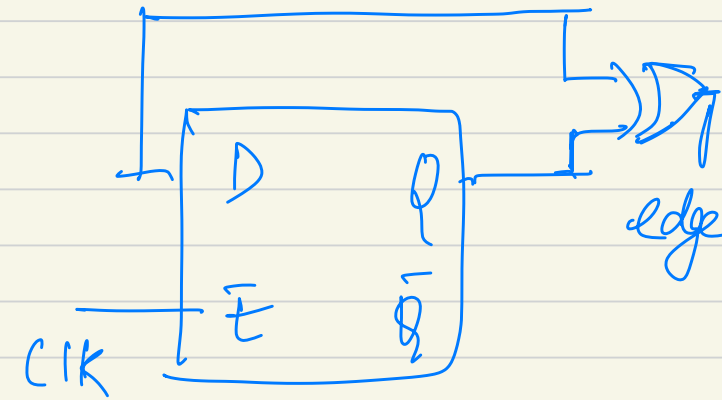
|   | D | $Q_n$ | $Q_{n+1}$ |
|---|---|-------|-----------|
| ↑ | 0 | X     | 0         |
| ↑ | 1 | X     | 1         |

$Q$  does not tell us when D is changing

it tells where D is making a transition.



if does not  
tell all input  
transitions

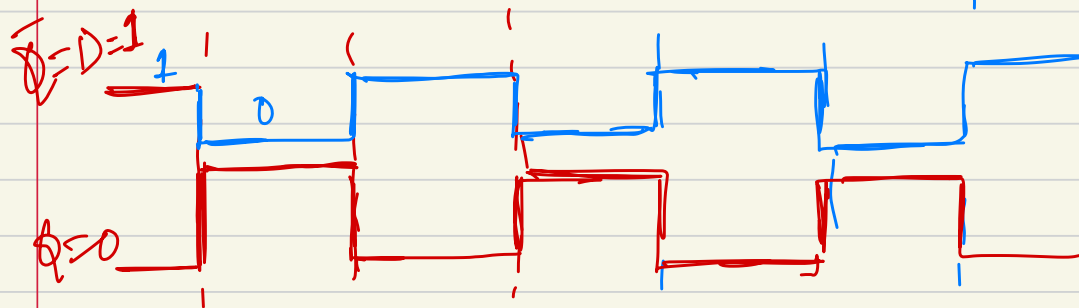
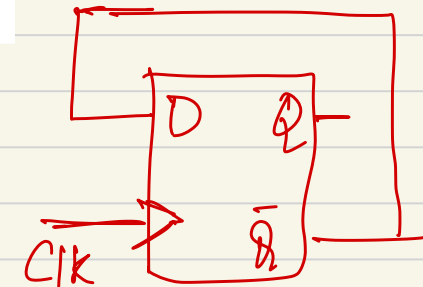
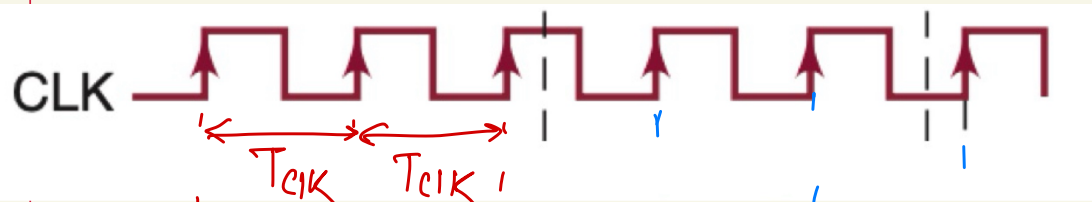


$E=0, CLK=0$

$$Q_{n+1} = Q_n$$

$E=1 \text{ or } CLK=1, Q_{n+1} = D$

Q2. You have a single positive-edge triggered D Flip-Flop. You wire the inverted output ( $\bar{Q}$ ) directly back into the D input. If you connect a clock signal (as shown in Question 1) to the clock pin, draw the output Q of the DFF? Comment on the frequency of CLK and obtained Q signal.



$$T_Q = 2 T_{CLK} \quad \therefore f_Q = f_{CLK} / 2 \quad (\text{Divide freq. by 2 circuit})$$



Q3. You are given the conversion table as shown here. To build the D Flip Flop using S-R FF,

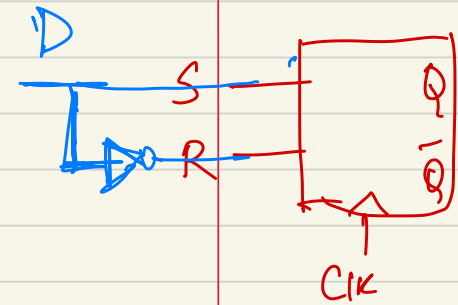
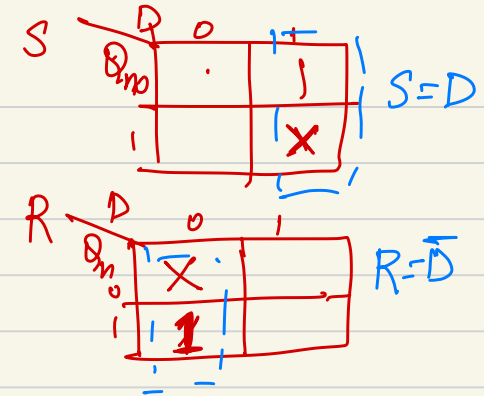
- Fill the required S and R inputs based on  $Q_n$  and  $Q_{n+1}$ . (i.e. Excitable Table for SR FF)
- Use K-maps to find S and R expression in terms of D and  $Q_n$ .
- Draw the resultant logic circuit using SR FF.

Input for K-map

| Input | Present state | Next state | Flip-flop inputs |   |
|-------|---------------|------------|------------------|---|
| D     | $Q_n$         | $Q_{n+1}$  | S                | R |
| 0     | 0             | 0          | 0                | X |
| 0     | 1             | 0          | 0                | 1 |
| 1     | 0             | 1          | 1                | 0 |
| 1     | 1             | 1          | X                | 0 |

outputs for K-map

State Table



| S | R | $Q_{n+1}$ |
|---|---|-----------|
| 0 | 0 | $Q_n$     |
| 0 | 1 | 0         |
| 1 | 0 | 1         |
| 1 | 1 | Invalid   |

Truth Table

| $Q_n$ | $Q_{n+1}$ | S | R |
|-------|-----------|---|---|
| 0     | 0         | 0 | X |
| 0     | 1         | 1 | 0 |
| 1     | 0         | 0 | 1 |
| 1     | 1         | X | 0 |

Excitation Table

3d: SR latch  $\rightarrow$  D latch with E

| E | D | $Q_n$ | $Q_{n+1}$ | S | R |
|---|---|-------|-----------|---|---|
| 0 | 0 | 0     | 0         | 0 | X |
| 0 | 0 | 1     | 1         | X | 0 |
| 0 | 1 | 0     | 0         | 0 | X |
| 0 | 1 | 1     | 1         | X | 0 |
| 1 | 0 | 0     | 0         | 0 | 1 |
| 1 | 0 | 1     | 0         | 0 | 1 |
| 1 | 1 | 0     | 1         | 1 | 0 |
| 1 | 1 | 1     | 1         | 1 | 0 |

for S

| $Q_n$ | 0 | 1 | 1 | 0 |
|-------|---|---|---|---|
| 0     | X | X |   |   |
| 1     |   | 1 | 1 |   |

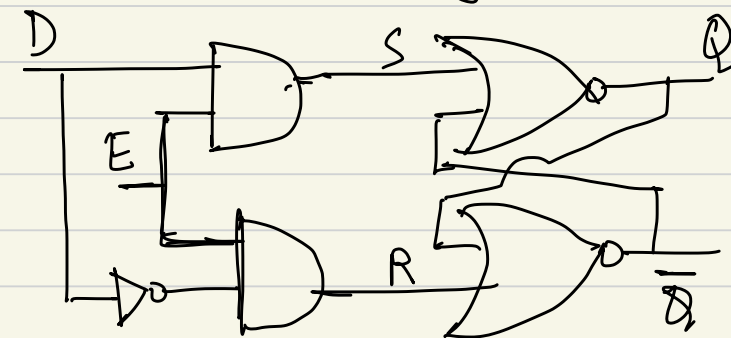
for R

| $Q_n$ | 0 | 1 | 1 | 0 |
|-------|---|---|---|---|
| 0     | X |   |   | X |
| 1     | 1 | 1 |   |   |

$S = ED$

$R = E\bar{D}$

SR latch Using NOR



Q4. You are designing a logic circuit for a simple cargo elevator that serves only two floors: Floor 0 (Ground) and Floor 1 (Top). State  $Q=0$  means the elevator is at Floor 0. State  $Q=1$  means the elevator is at Floor 1. There is a single "Move" button, labeled  $M$ . If  $M=0$ , the elevator stays where it is. If  $M=1$ , the elevator moves to the *other* floor.

- Draw the state transition diagram (circles for states, arrows for transitions based on  $M$ ).
- Fill out the State Transition Table.
- Derive the Boolean logic equation for the Flip-Flop input  $D$  in terms of current state  $Q$  and input  $M$ . What basic logic gate does this represent?

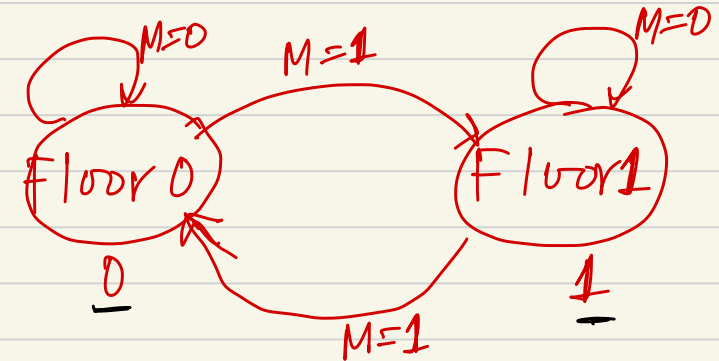
State Table

| $M$ | $Q_0(t)$ | $Q_0(t+1)$ | $D_0$ |
|-----|----------|------------|-------|
| 0   | 0        | 0          | 0     |
| 0   | 1        | 1          | 1     |
| 1   | 0        | 1          | 1     |
| 1   | 1        | 0          | 0     |

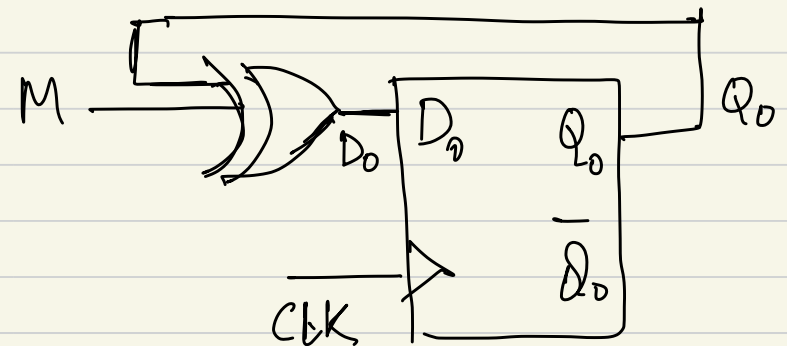
| $M$ | $Q_0$ | $D_0$ |
|-----|-------|-------|
| 0   | 0     | 0     |
| 0   | 1     | 1     |
| 1   | 0     | 1     |
| 1   | 1     | 0     |

$$D_0 = M \oplus Q_0$$

Move  $\rightarrow$  External Input ( $M$ )



only one DFF needed.



Q5. Design a simple 2-bit synchronous counter that continuously counts upward: 00 → 01 → 10 → 11 → 00, using D Flip-Flops.

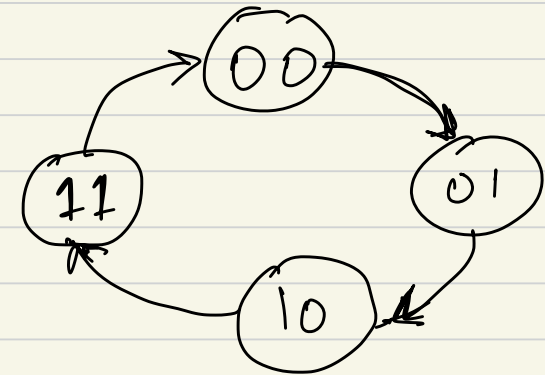
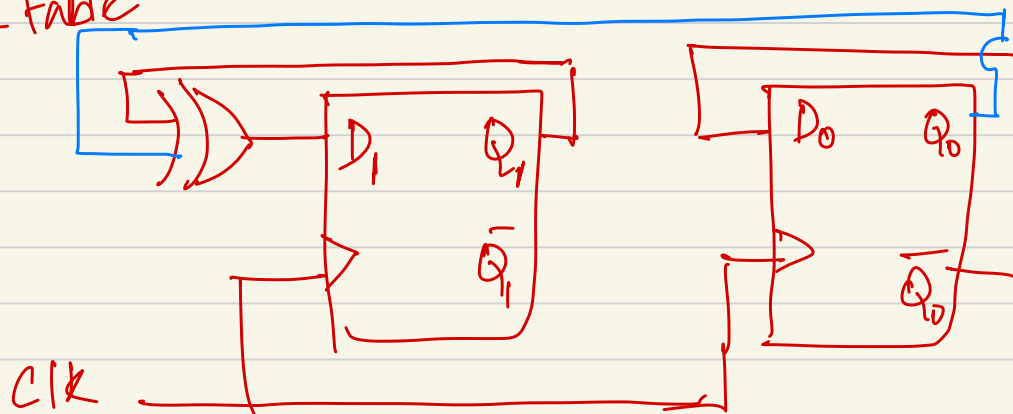
- Write the State Transition Table showing Current State ( $Q_1, Q_0$ ) and Next State ( $D_1, D_0$ ).
- Find the simplified Boolean logic equation for  $D_0$ .
- Find the simplified Boolean logic equation for  $D_1$ .

no external input

Two DFF.

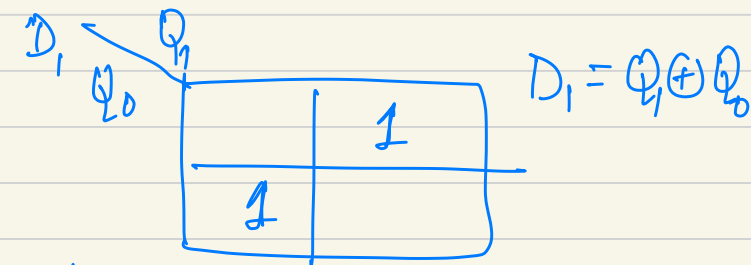
| $Q_1(t)$ | $Q_0(t)$ | $Q_1(t+1)$ | $Q_0(t+1)$ | $D_1$ | $D_0$ |
|----------|----------|------------|------------|-------|-------|
| 0        | 0        | 0          | 1          | 0     | 1     |
| 0        | 1        | 1          | 0          | 1     | 0     |
| 1        | 0        | 1          | 1          | 1     | 1     |
| 1        | 1        | 0          | 0          | 0     | 0     |

State table

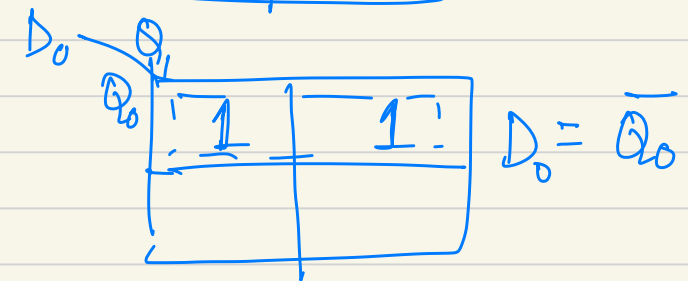


$$D_1 = f(Q_1, Q_0)$$

$$D_0 = f(Q_1, Q_0)$$



$$D_1 = Q_1 \oplus Q_0$$



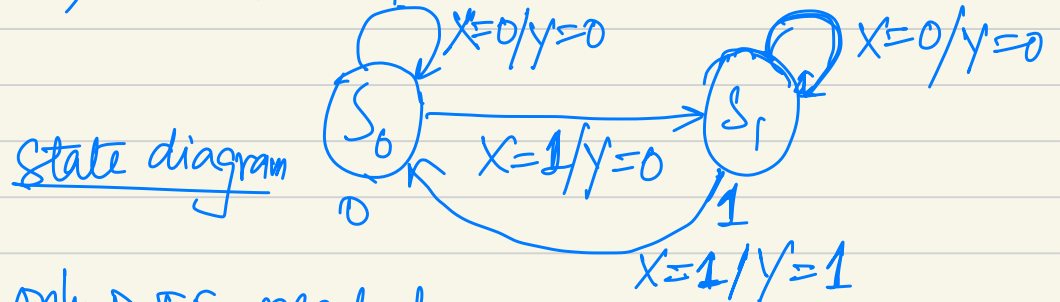
$$D_0 = \overline{Q_0}$$

Q6. The campus has installed a new automated Chai dispenser. A cup of chai costs exactly 20 Rupees. The machine only accepts 10-Rupee coins.

A. To save hardware costs, your institute decides to install an "instant" dispenser. The exact moment the second 10-Rupee coin is inserted, it immediately pours the chai and resets. It never "waits" with 20 Rupees inside. Input X: X=1 for exactly one clock cycle when a 10-Rupee coin is inserted (X=0 otherwise). Output Y: Y=1 triggers the dispenser to pour the chai (Y=0 keeps it locked). Use exactly one D Flip-Flop. State 0 (Q=0) means 0 Rupees inserted. State 1 (Q=1) means 10 Rupees inserted.

1. Draw the State Transition Diagram (using arrows labeled X/Y).
2. Fill out the State Transition Table.
3. Derive the simplified logic equations for the Next State (D) and Output (Y).
4. Sketch the final gate-level circuit.

X: External input (10 Rupee coin)



only D FF needed.

Y = 1, When Chai is poured.

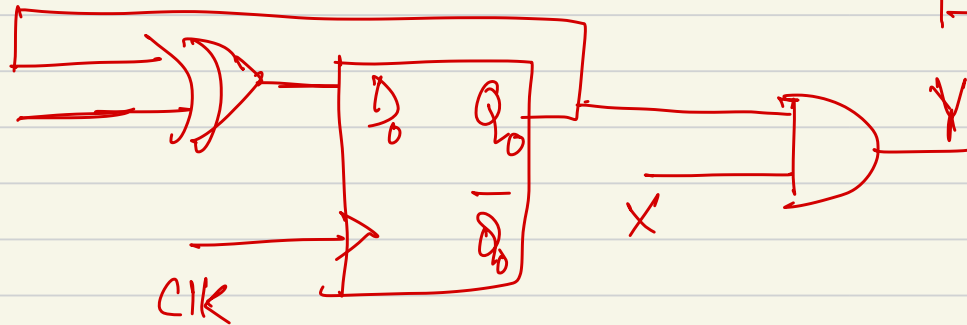
| X | Q <sub>0</sub> (t) | Q <sub>0</sub> (t+1) | Y | D <sub>0</sub> |
|---|--------------------|----------------------|---|----------------|
| 0 | 0                  | 0                    | 0 | 0              |
| 0 | 1                  | 1                    | 0 | 1              |
| 1 | 0                  | 1                    | 0 | 1              |
| 1 | 1                  | 0                    | 1 | 0              |

| D <sub>0</sub> | X | Q <sub>0</sub> |
|----------------|---|----------------|
| 0              | 0 | 0              |
| 1              | 0 | 1              |
| 0              | 1 | 0              |
| 1              | 1 | 1              |

$$D_0 = Q_0 \oplus X$$

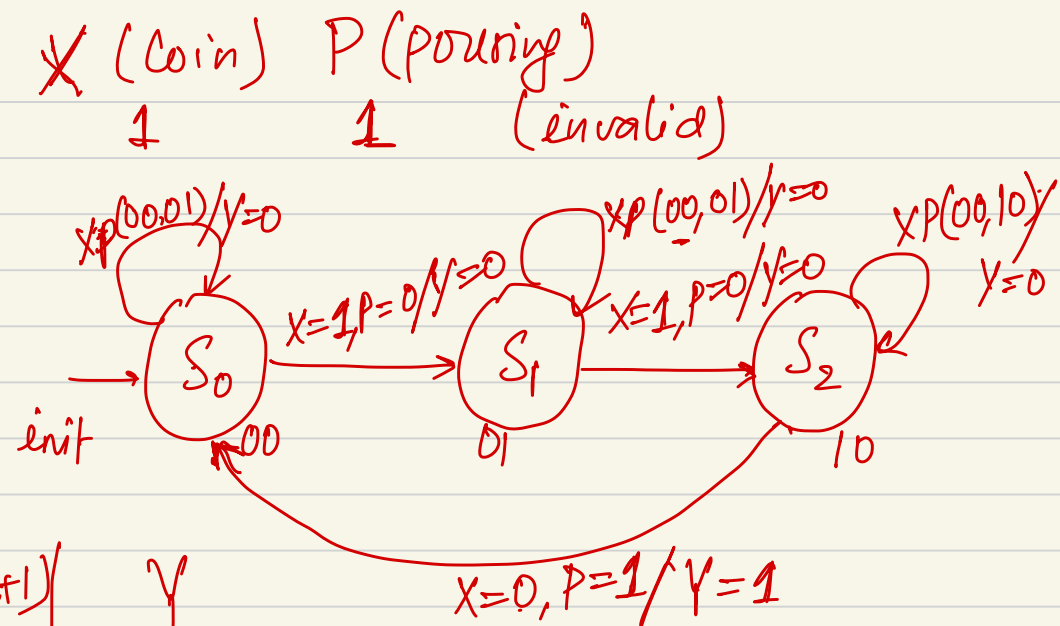
| Y | X | Q <sub>0</sub> |
|---|---|----------------|
| 0 | 0 | 0              |
| 0 | 1 | 0              |
| 1 | 0 | 1              |
| 1 | 1 | 1              |

$$Y = Q_0 X$$



B. Students complained that the "instant" dispenser from Part A spilled chai on their hands because it poured too fast! You are tasked with upgrading the machine. The machine must now hold the 20 Rupees and wait for the user to press a button before pouring. Input  $X=1$  when a 10-Rupee coin is inserted,  $P=1$  when the "Pour" button is pressed. (Assume  $X$  and  $P$  are never pressed at the exact same time). Output  $Y$ :  $Y=1$  pours the chai. The machine now requires three states:  $S_0$  (00): 0 Rupees,  $S_1$  (01): 10 Rupee,  $S_2$  (10): 20 Rupees (Ready to pour), and  $S_3$  (11): Invalid / Don't Care.

1. Draw the new State Transition Diagram. Include states  $S_0$ ,  $S_1$ , and  $S_2$ . Show what happens if  $X$  is inserted, and what happens if  $P$  is pressed. (Hint: If  $P$  is pressed while in  $S_0$  or  $S_1$ , the machine should just stay in that state).
2. Create the State Transition Table. There are 2 Flip-Flops ( $Q_1$ ,  $Q_0$ ) and 2 External Inputs ( $X$ ,  $P$ ).
3. Derive the logic equations for the Next States ( $D_1$ ,  $D_0$ ) and Output ( $Y$ ).



input

| X | P | $Q_1(t)$ | $Q_0(t)$ | $D_1 = Q_1(t+1)$ | $D_0 = Q_0(t+1)$ | Y |
|---|---|----------|----------|------------------|------------------|---|
| 0 | 0 | 0        | 0        | 0                | 0                | 0 |
| 0 | 0 | 0        | 1        | 0                | 1                | 0 |
| 0 | 0 | 1        | 0        | 1                | 0                | 0 |
| 0 | 0 | 1        | 1        | X                | X                | X |
| 0 | 1 | 0        | 0        | 0                | 0                | 0 |
| 0 | 1 | 0        | 1        | 0                | 1                | 0 |
| 0 | 1 | 1        | 0        | 1                | 0                | 1 |
| 0 | 1 | 1        | 1        | X                | X                | X |
| 1 | 0 | 0        | 0        | 0                | 0                | 0 |
| 1 | 0 | 0        | 1        | 0                | 1                | 0 |
| 1 | 0 | 1        | 0        | 1                | 0                | 0 |
| 1 | 0 | 1        | 1        | 1                | 1                | 0 |

k-Map for  
 $D_1$ ,  $D_0$  &  $Y$   
 $= f(X, P, Q_1, Q_0)$

|   |   |   |   |
|---|---|---|---|
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 |

|   |   |   |
|---|---|---|
| X | X | X |
| X | X | X |
| X | X | X |
| X | X | X |
| X | X | X |

$D_1$

| XP | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 |    |    | X  |    |
| 01 |    |    | X  | 1  |
| 11 | X  | X  | X  | X  |
| 10 | 1  |    | X  | 1  |

Y

$$D_1 = X Q_0 + \bar{P} Q_1$$

$$D_0 = \bar{X} Q_0 + X \bar{Q}_0 \bar{Q}_1$$

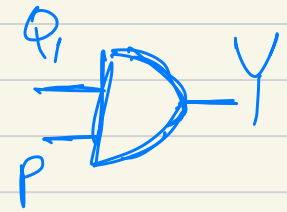
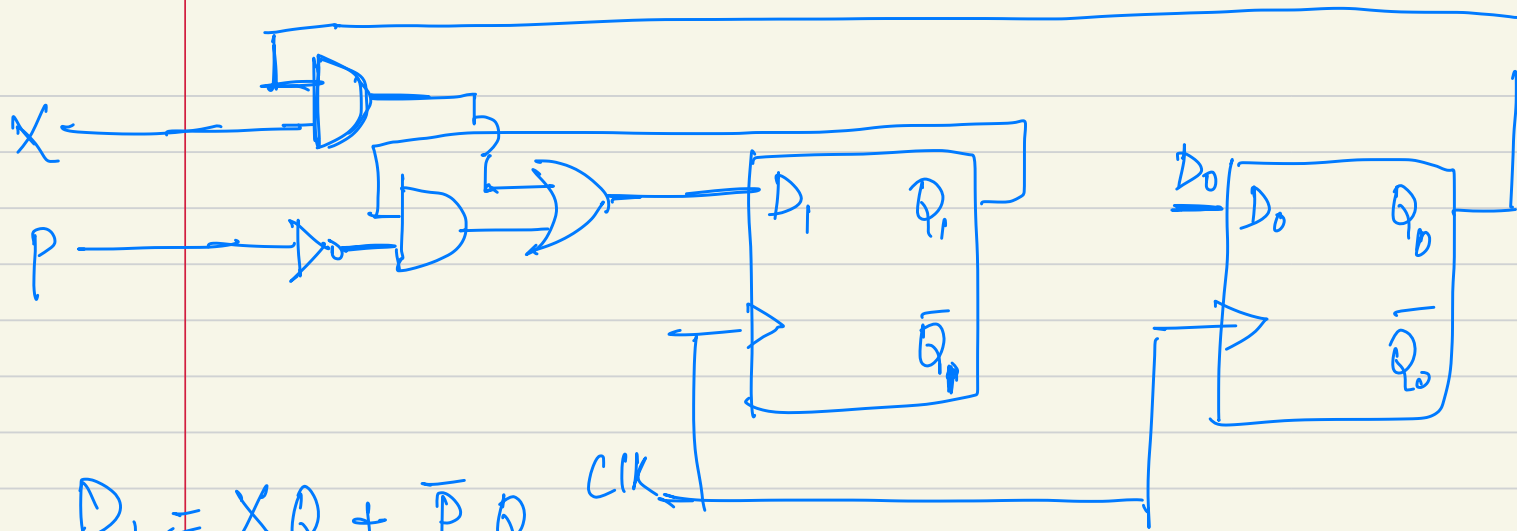
$$Y = Q_1 P$$

$D_0$

| XP | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 |    |    | X  | 1  |
| 01 | 1  | 1  | X  |    |
| 11 | X  | X  | X  | X  |
| 10 |    |    | X  |    |

Y

| XP | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 |    |    | X  |    |
| 01 |    |    | X  |    |
| 11 | X  | X  | X  | X  |
| 10 |    | 1  | X  |    |



$$D_1 = X Q_0 + \bar{P} Q_1$$

$$D_0 = \bar{X} Q_0 + X \bar{Q}_0 \bar{Q}_1$$

$$Y = Q_1 P$$

